Soru Çözümü 1

30 Aralık 2021 Perşembe 11:58

İntegral Soru...

SORU $\int (x+1)\left(x^2+2x-1\right)^{10}\,dx$ integralini hesaplayınız.

$$x^2+2x-1 = u$$

$$(2x+2)\,dx = du$$

$$2(x+1)\,dx = du$$

$$(x+1)\,dx = \frac{1}{2}\,du$$

$$= \frac{1}{2}\int u^{10}\,du = \frac{1}{2}\left(\frac{u^{11}}{11}\right) + C$$

$$= \frac{1}{22}\left(x^2+2x-1\right)^{11} + C$$

$x^2+2x+5=t$

$(2x+2)dx=dt$

SORU $\int \frac{2x+3}{x^2+2x+5}dx$ integralini hesaplayınız.

$$=\int \frac{2x+2+1}{x^2+2x+5}dx$$

$$=\int \frac{2x+2}{x^2+2x+5}dx+\int \frac{dx}{\underbrace{x^2+2x+5}_{(x+1)^2+4}}$$

$$\int \frac{dx}{a^2+x^2}=\frac{1}{a}\operatorname{arctg}\frac{x}{a}+c$$

$x^2+2x+5=t$

$(2x+2)dx=dt$

$$=\int \frac{dt}{t}=\ln t+c$$

$$=\ln(x^2+2x+5)+c$$

$$\int \frac{dx}{(x+1)^2+4}=\int \frac{du}{u^2+4}=\frac{1}{2}\operatorname{arctg}\frac{u}{2}+C$$

$x+1=u$

$dx=du$

$$=\frac{1}{2}\operatorname{arctg}\left(\frac{x+1}{2}\right)+c$$

$$=\ln(x^2+2x+5)+\frac{1}{2}\operatorname{arctg}\left(\frac{x+1}{2}\right)+C \;//$$

SORU $\int \frac{\sin^{2/3} x}{\cos^{8/3} x} dx$ integralini hesaplayınız.

$$= \int \frac{\sin^{2/3} x}{\cos^{2/3} x \cdot \cos^{2} x} dx = \int \frac{\tan^{2/3} x}{\cos^2 x} \cdot dx = \int \tan^{2/3} x \cdot \sec^2 x dx$$

$\tan x = t$

$\sec^2 x \, dx = dt$

$$= \int t^{2/3} dt$$

$$= \frac{t^{5/3}}{\frac{5}{3}} + C$$

$$= \frac{3}{5} \cdot \tan^{5/3} x + C$$

SORU $\int \frac{\sqrt{x^2+a^2}}{x}dx$ integralini hesaplayınız.

$x=a\tan t$, $dx=\sec^2 t\,dt$

$x^2+a^2=a^2\tan^2 t+a^2$

$=a^2(\tan^2 t+1)$

$=a^2\sec^2 t$?

$$=\int \frac{\sqrt{x^2+a^2}}{x}\cdot\frac{x}{x}\,dx$$

$$=\int \frac{\sqrt{x^2+a^2}}{x^2}\cdot x\,dx = \int \frac{\sqrt{t^2}}{t^2-a^2}\cdot t\,dt = \int \frac{t^2}{t^2-a^2}\,dt$$

$x^2+a^2=t^2 \Rightarrow t=\sqrt{x^2+a^2}$

$2x\,dx=2t\,dt$

$x\,dx=t\,dt$

$$\begin{array}{r|l} t^2 & t^2-a^2 \\ -\ t^2-a^2 & 1 \\ \hline a^2 & \end{array}$$

$$\frac{t^2}{t^2-a^2}=1+\frac{a^2}{t^2-a^2}$$

$$=\int dt+\int \frac{a^2}{t^2-a^2}\,dt$$

$$\int \frac{dx}{x^2-a^2}$$

$$\frac{1}{x^2-a^2}=\frac{A}{x-a}+\frac{B}{x+a}=\frac{1}{2a}\int\frac{dx}{x-a}-\frac{1}{2a}\int\frac{dx}{x+a}$$

$1=A(x+a)+B(x-a)$

$x=a,\quad 1=A\cdot 2a$

$A=\frac{1}{2a}$

$x=-a,\quad 1=B\cdot(-2a)$

$B=-\frac{1}{2a}$

$$=\frac{1}{2a}\ln(x-a)-\frac{1}{2a}\ln(x+a)$$

$$=\frac{1}{2a}\left(\ln(x-a)-\ln(x+a)\right)$$

$$=\frac{1}{2a}\ln\left(\frac{x-a}{x+a}\right)+c$$

$$=t+a^2\cdot\frac{1}{2a}\ln\left|\frac{t-a}{t+a}\right|+c$$

$$=\sqrt{x^2+a^2}+\frac{a}{2}\ln\left|\frac{\sqrt{x^2+a^2}-a}{\sqrt{x^2+a^2}+a}\right|+c$$

SORU $\int \frac{\cos^3 x+\cos^5 x}{\sin^2 x+\sin^4 x}dx$ integralini hesaplayınız.

$$=\int \frac{(\cos^2 x+\cos^4 x)\cos x}{\sin^2 x+\sin^4 x}\,dx$$

$$=\int \frac{(1-\sin^2 x)+(1-\sin^2 x)^2}{\sin^2 x+\sin^4 x}\cos x\,dx \quad , \quad \sin x=t,\ \cos x\,dx=dt$$

$$=\int \frac{(1-t^2)+(1-t^2)^2}{t^2+t^4}\,dt$$

$$=\int \frac{(1-t^2)+(t^4-2t^2+1)}{t^2+t^4}\,dt$$

$$=\int \frac{t^4-3t^2+2}{t^4+t^2}\,dt$$

$$\begin{array}{r|l} t^4-3t^2+2 & t^4+t^2 \\ t^4+t^2 & 1 \\ \hline -4t^2+2 & \end{array}$$

$$= \int \frac{t^4 - 3t^2 + 2}{t^4 + t^2}\, dt$$

$$\begin{array}{l} t^4 + t^2 \\ \hline -4t^2 + 2 \end{array}$$

$$= \int 1\, dt + \int \frac{-4t^2 + 2}{t^4 + t^2}\, dt$$

$$\frac{-4t^2+2}{t^2(1+t^2)} = \frac{A}{t} + \frac{B}{t^2} + \frac{Ct+D}{1+t^2}$$

$$t(1+t^2) \qquad 1+t^2 \qquad t^2$$

$$-4t^2 + 2 = At^3 + At + B + Bt^2 + Ct^3 + Dt^2$$

$A + C = 0 \quad \Rightarrow C = 0$

$B + D = -4 \quad D = -6$

$A = 0$

$B = 2$

$$= \int dt + 2\int \frac{dt}{t^2} - 6\int \frac{dt}{1+t^2}$$

$$= t - \frac{2}{t} - 6 \operatorname{arctg} t + c$$

$$= \sin x - 2 \operatorname{cosec} x - 6 \operatorname{arctg}(\sin x) + c$$

SORU $\int \frac{x^2dx}{\sqrt{8x-4x^2}}$ integralini hesaplayınız.

$= \frac{1}{2} \int \frac{x^2 dx}{\sqrt{2x-x^2}}$

$-(x^2+2x)$

$1-(x+1)^2$

Yol gösterme

ödev

SORU $\int x\sin^2 2x\,dx$ integralini hesaplayınız.

$\int \sin^2 x\,dx\ ,\ \int \cos^2 x\,dx$

$\cos 2x = \cos^2 x - \sin^2 x$

$= 1 - 2\sin^2 x$

$= 2\cos^2 x - 1$

$* \sin^2 x = \dfrac{1-\cos 2x}{2}$

$\cos^2 x = \dfrac{1+\cos 2x}{2}$

$$= \int x \cdot \left(\frac{1-\cos 4x}{2}\right) dx$$

$$= \frac{1}{2}\int (x - x\cos 4x)\,dx$$

$$= \frac{1}{2}\int x\,dx - \frac{1}{2}\underbrace{\int x\cos 4x\,dx}$$

$x = u\ ,\ \cos 4x\,dx = dv$

$dx = du\ ,\ \dfrac{\sin 4x}{4} = v$

$$\int x\cos 4x\,dx = \frac{x\sin 4x}{4} - \frac{1}{4}\int \sin 4x\,dx$$

$$= \frac{x\sin 4x}{4} + \frac{1}{16}\cos 4x + c$$

$\int x\sin^2 2x\,dx = ?$

$x = u\ ,\ \underbrace{\sin^2 2x\,dx} = dv$

$$= \frac{x^2}{4} - \frac{1}{8}x\sin 4x - \frac{1}{32}\cos 4x + c \quad //$$

SORU $\int \sqrt{\frac{x}{1-x^3}}dx$ integralini hesaplayınız.

$$= \int \frac{\sqrt{x}}{\sqrt{1-x^3}}dx = \frac{2}{3}\int \frac{dt}{\sqrt{1-t^2}} = \frac{2}{3}\arcsin t + C$$

$$= \frac{2}{3}\arcsin\left(x^{3/2}\right) + C$$

$$x^{\frac{3}{2}} = t$$

$$\frac{3}{2}\cdot\sqrt{x}\,dx = dt$$

$$\sqrt{x}\,dx = \frac{2}{3}dt$$